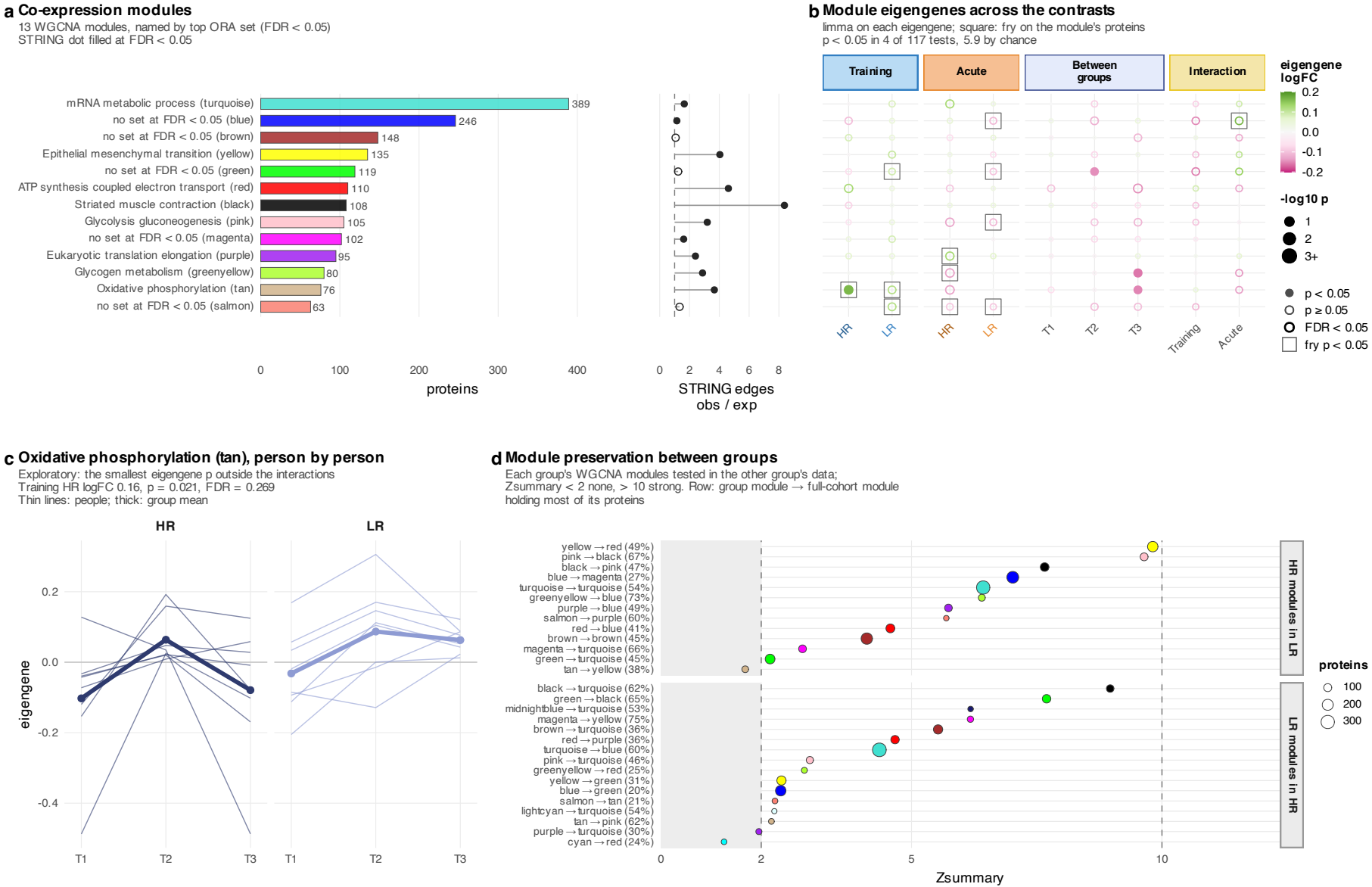
a Co-expression modules
13 WGCNA modules, named by top ORA set (FDR < 0.05)
STRING dot filled at FDR < 0.05
mRNA metabolic process (turquoise) 389
no set at FDR < 0.05 (blue) 246
no set at FDR < 0.05 (brown) 148
Epithelial mesenchymal transition (yellow) 135
no set at FDR < 0.05 (green) 119
ATP synthesis coupled electron transport (red) 110
Striated muscle contraction (black) 108
Glycolysis gluconeogenesis (pink) 105
no set at FDR < 0.05 (magenta) 102
Eukaryotic translation elongation (purple) 95
Glycogen metabolism (greenyellow) 80
Oxidative phosphorylation (tan) 76
no set at FDR < 0.05 (salmon) 63
0 100 200 300 400
proteins
0 2 4 6 8
STRING edges obs / exp
b Module eigengenes across the contrasts
limma on each eigengene; square: fry on the module's proteins
p < 0.05 in 4 of 117 tests, 5.9 by chance
Training
Acute
Between groups
Interaction
HR LR HR LR T1 T2 T3 Training Acute
eigengene logFC
0.2 0.1 0.0 -0.1 -0.2
-log10 p
1 2 3+
p < 0.05
p ≥ 0.05
FDR < 0.05
fry p < 0.05
c Oxidative phosphorylation (tan), person by person
Exploratory: the smallest eigengene p outside the interactions
Training HR logFC 0.16, p = 0.021, FDR = 0.269
Thin lines: people; thick: group mean
HR
LR
eigengene
0.2 0.0 -0.2 -0.4
T1 T2 T3 T1 T2 T3
d Module preservation between groups
Each group's WGCNA modules tested in the other group's data;
Zsummary < 2 none, > 10 strong. Row: group module → full-cohort module holding most of its proteins
yellow → red (49%)
pink → black (67%)
black → pink (47%)
blue → magenta (27%)
turquoise → turquoise (54%)
greenyellow → blue (73%)
purple → blue (49%)
salmon → purple (60%)
red → blue (41%)
brown → brown (45%)
magenta → turquoise (66%)
green → turquoise (45%)
tan → yellow (38%)
black → turquoise (62%)
green → black (65%)
midnightblue → turquoise (53%)
magenta → yellow (75%)
brown → turquoise (36%)
red → purple (36%)
turquoise → blue (60%)
pink → turquoise (46%)
greenyellow → red (25%)
yellow → green (31%)
blue → green (20%)
salmon → tan (21%)
lightcyan → turquoise (54%)
tan → pink (62%)
purple → turquoise (30%)
cyan → red (24%)
HR modules in LR
LR modules in HR
proteins
100 200 300
0 2 5 10
Zsummary